Aarnav Singh

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Professional Summary

Final-year B.Tech CSE student (Cybersecurity specialization) with hands-on experience in risk detection, anomaly analysis, and policy-driven abuse mitigation across ML-based security systems. Co-author of a research paper under review at the Journal of Information Security and Applications (Elsevier), demonstrating how context-aware policy engines can reduce evasion/bypass rates from 65% to 2.2% — directly analogous to detecting and mitigating adversarial abuse patterns at scale.

Education

SRM University India
B.Tech in Computer Science & Engineering with specialization in Cybersecurity *2022 – 2026*
- GPA: 7.74/10

Experience

HCLTech – Intern May 2024 – July 2024
Noida, India
- Built foundational skills in enterprise network security and perimeter security – including firewalls, network segmentation, and threat mitigation strategies.
- Supported identification and mitigation of network security risks, contributing to incident documentation and response workflows.
- Applied Python for DevSecOps automation, including security tooling configuration and codebase review to support automated security workflows.

Projects

Adversarial Attack Detection in Zero-Trust Networks Jan 2026
- *GitHub:* github.com/Aarnav-Singh/adversarial-ml-security-framework
- Built a six-layer Zero-Trust network security system in Python/PyTorch combining ML-based intrusion detection with a context-aware policy engine.
- Designed a five-decision policy engine (ALLOW / DENY / STEP_UP_AUTH / RATE_LIMIT / ISOLATE) that made context-aware, high-stakes decisions in real time, reducing abuse/evasion bypass rates from 65% to 2.2% without requiring model retraining.
- Analyzed gaps in existing detection policy and designed and implemented solutions to close them, validated through a 5-seed evaluation across two large-scale real-world datasets (UNSW-NB15, CICIDS-2017).
- Co-authored a research paper based on this work, currently under review at the Journal of Information Security and Applications (Elsevier).

Hybrid Behavioral Intrusion Detection System Feb 2026
- *GitHub:* github.com/Aarnav-Singh/Hybrid-Behavioral-Intrusion-Detection-System
- Designed and developed a full-stack hybrid intrusion detection system combining rule-based detection with Isolation Forest-based anomaly detection for behavioral threat analysis.
- Built a real-time React dashboard using WebSockets and REST APIs for live threat visualization and alerting.
- Integrated Elasticsearch for centralized log indexing and threat-event correlation, and deployed Prometheus for monitoring detection latency and system performance.
- Containerized the full system using Docker to simulate a scalable deployment environment aligned with Zero-Trust and modern security monitoring principles.

Technical Skills

Networking & Zero-Trust Security: TCP/IP, DNS, Routing & Switching, Network Segmentation, Firewalls, VPN, Zero-Trust Architecture (ZTNA), SASE, Secure SD-WAN, NAC

Analysis & Reporting: Data-driven risk analysis, root-cause investigation, technical documentation, cross-functional collaboration

Adversarial ML & Anomaly Detection: Adversarial Machine Learning, Isolation Forest, Anomaly Detection, Feature Engineering, ROC Curve / Confusion Matrix Analysis

Security Operations & Detection: IDS/IPS, Log Analysis & Correlation, Threat Detection & Cyber Threat Intelligence (CTI), Vulnerability Assessment, Incident Response, MITRE ATT&CK Framework

Tools & Platforms: Wireshark, Nmap, Nessus, Metasploit, FortiGate, Kali Linux, Elasticsearch, Web Application Firewalls, Git/GitHub

Cloud & Development: Google Cloud Platform (GCP), Microsoft Azure, AWS, IAM, DevSecOps, Linux, Windows, Python, PyTorch, Docker, React.js, REST APIs, WebSockets

Certifications

Google Cloud Cybersecurity Certificate

Fortinet NSE 3 Certified in Cybersecurity

Fortinet NSE 2 Certified in Cybersecurity

Fortinet NSE 1 Certified in Cybersecurity

Cisco – Networking Basics

NPTEL – Advanced Computer Networks

Deloitte Australia (Forage) – Cyber Job Simulation

Leadership & Activities

Volunteer – Purvanchal Vikas Foundation June 2024 – July 2024
Noida, India
- Supported outreach initiatives and beneficiary enrollment programs.

Core Member & Event Host – Gamer's Creed Esports Club
Ghaziabad, India
- Hosted and coordinated college-level esports events.